

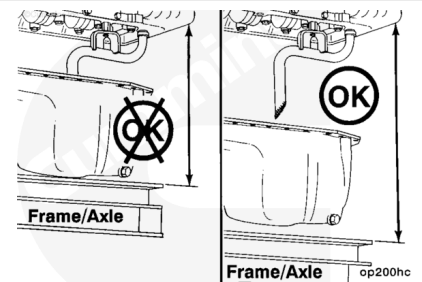
Trainings ▾

Remove

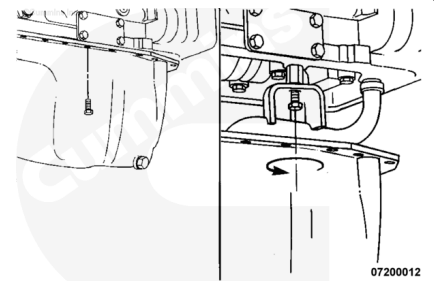
Front Sump Pan

Front sump oil pans have a block-mounted suction tube, which requires approximately 610 mm [24 in] clearance between the vehicle frame or axle and the cylinder block oil pan mounting flange to remove the oil pan.

If an application will **not** allow the oil pan to be lowered that much, the suction tube **must** be removed from the cylinder block before the oil pan can be removed from the chassis.

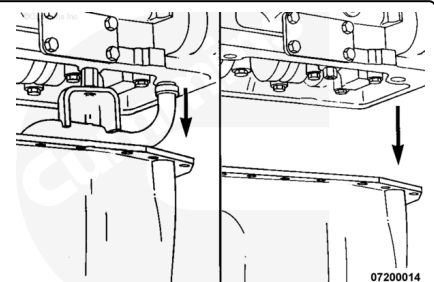


Remove the oil pan mounting capscrews. Lower the oil pan far enough to allow one hand to reach inside the engine and remove the suction tube mounting capscrew.



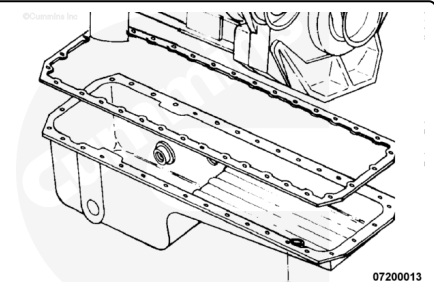
Remove the suction tube from the cylinder block. Let it drop into the oil pan.

Remove the oil pan from the chassis.



Rear Sump Pan

Remove the capscrews and oil pan.



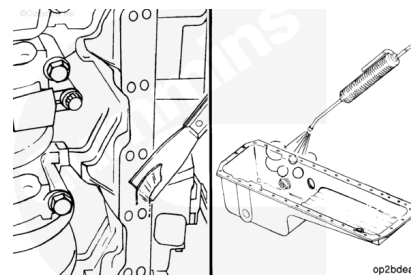
Clean and Inspect for Reuse

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



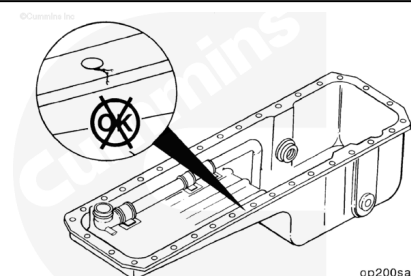
Remove all gasket material from the cylinder block and oil pan surfaces.

Steam clean the oil pan. Dry with compressed air.

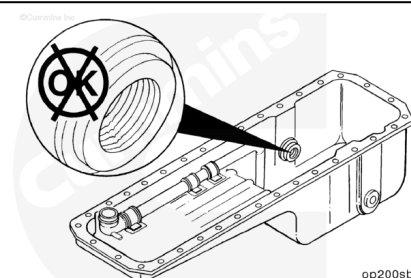
Inspect the oil pan, suction tube, and tube braces for cracks or damage.

Note : If cracks are found, replace the damaged part. Do not attempt to repair the oil pan by welding.

Inspect the oil transfer tube. Refer to Procedure 007-040 in Section 7. (</qs3/pubsys2/xml/en/procedures/101/101-007-040-tr.html>)



Inspect the threaded holes for damage. Damaged threads can be chased with the following taps.



Oil Pan Threaded Holes

Thread Location	Tap Size
Temperature Gauge	1/2 in NPTF
Oil Sump Heater	1 in NPTF
Oil Drain	1-18 NS-3B

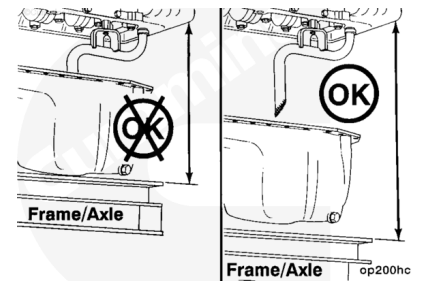
Note : If the threaded holes can not be repaired with the taps listed above, the oil pan must be replaced.

Install

Front Sump Pan

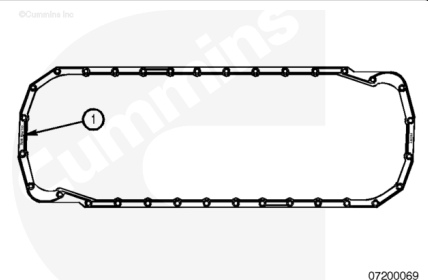
On front sump oil pans, if the application does **not** allow approximately 610 mm [24 in] clearance between vehicle frame or axle and the cylinder block oil pan mounting surface during installation, the suction tube and oil pan will need to be installed together as an assembly. Installation requires approximately this much clearance to maneuver the oil pan over the frame or axle and under the block-mounted suction tube when installing the pieces separately. Follow all the steps as outlined on the following pages.

If the application does allow approximately 610 mm [24 in] clearance, install the suction tube first by itself, then the oil pan.

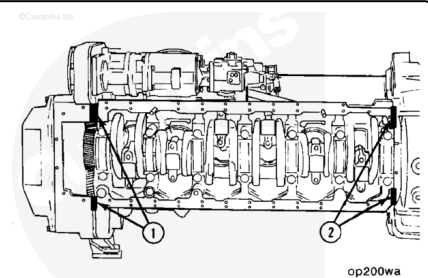


Assemble a new gasket onto the oil pan.

If using the silicone beaded gasket, place the oil pan gasket on the oil pan mounting flange with the words OIL PAN SIDE (1) facing the oil pan.

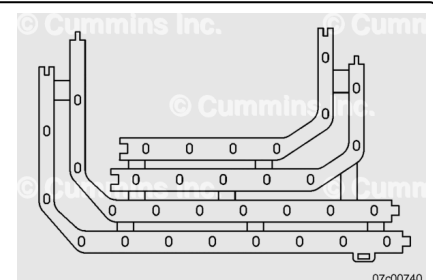


Apply a 2 mm [0.08 in] bead of sealant, Part Number 3164067, to the oil pan mounting surfaces at the cylinder block to gear housing joint (1) and flywheel housing to cylinder block joint (2).



The replacement lubricating oil pan gasket may be in be a four piece segmented lubricating oil pan gasket rather than a single piece lubricating oil pan gasket.

The replacement lubricating oil pan gasket will need to be separated into four pieces and assembled on the lubricating oil pan flange to make a complete lubricating oil pan gasket.



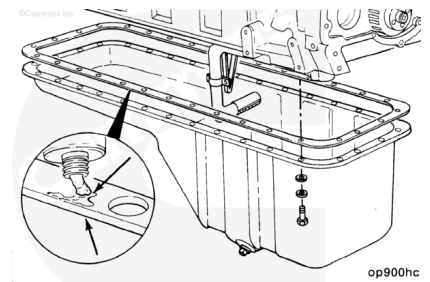
Apply a 2 mm [0.08 in] bead of sealant, Part Number 3164067, to the lubricating oil pan flange. Sealant must be applied around the circumference of the oil pan mounting holes.

Lay the lubricating oil pan gasket(s) onto the lubricating oil pan flange.

For segmented lubricating oil pan gaskets, make sure the lubricating oil pan gasket segment joints are properly aligned.

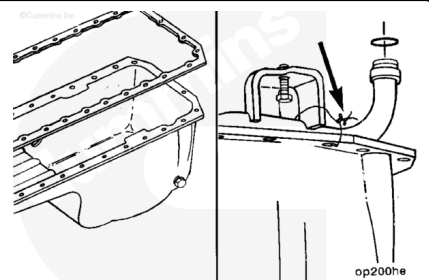
Apply a 2 mm [0.08 in] bead of sealant, Part Number 3164067, to the top of the new lubricating pan gasket.

For segmented lubricating oil pan gaskets, make sure the sealant fills the lubricating oil pan gasket segment joints.



Install a new o-ring to the oil suction tube. Use clean 15W-40 engine oil to lubricate the o-ring.

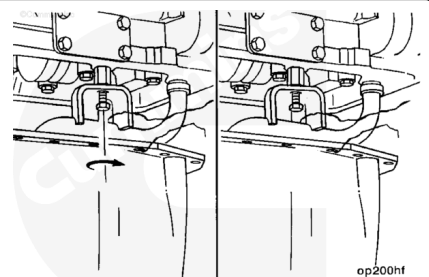
Attach the suction tube to the oil pan mounting flange with a piece of wire. This will hold the suction tube and oil pan together as one assembly until the tube is fastened to the cylinder block.



Maneuver the assembly into place under the engine in the chassis.

Align the suction tube with the cylinder block suction tube hole. Reach inside the engine, and start the capscrew into the mounting hole to hold the suction tube in place.

After the capscrew is started in the mounting hole by several threads, the wire holding the suction tube to the oil pan can be removed and the oil pan can be lowered to allow more hand clearance to complete the suction tube installation.

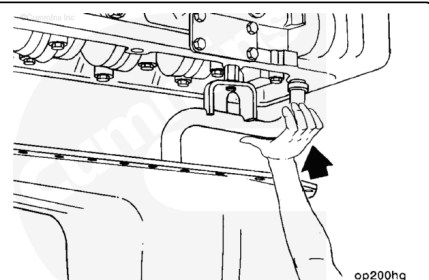


⚠ CAUTION ⚠

Do not use the capscrew to pull the suction tube into place during installation. Doing so will cause misalignment of the suction tube, which can cause damage to the engine.

Reach inside the engine. Push the suction tube completely up into place in the cylinder block suction tube mounting hole by hand.

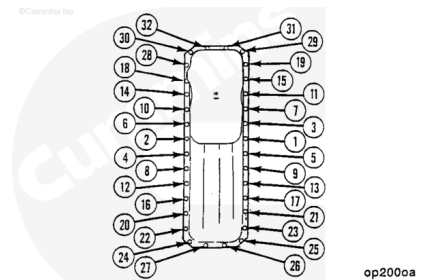
Complete the suction tube retaining capscrew installation.



Torque Value: 61 n•m [45 ft-lb]

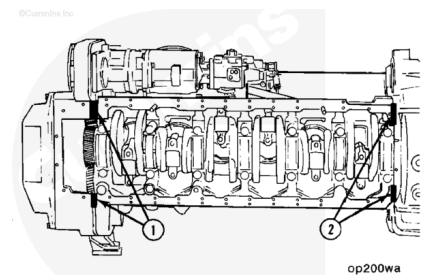
Install the oil pan. Tighten the capscrews in the sequence shown.

Torque Value: 54 n•m [40 ft-lb]



Rear Sump Pan

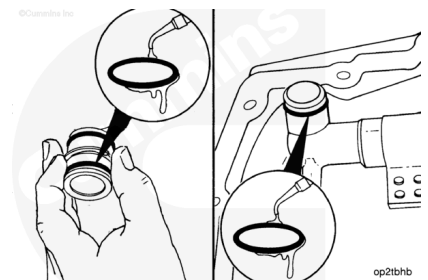
Apply a 2 mm [0.08 in] bead of sealant, Part Number 3164067, to the oil pan mounting surfaces at the cylinder block to gear housing joint (1) and flywheel housing to cylinder block joint (2).



Install new o-rings on the oil transfer tube.

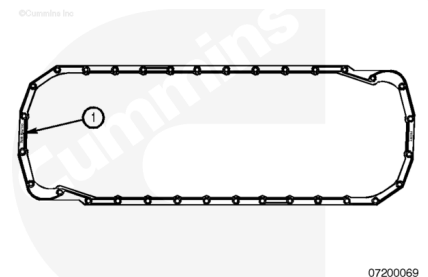
Use clean 15W-40 oil to lubricate the o-rings.

Install the transfer tube in the oil pan.



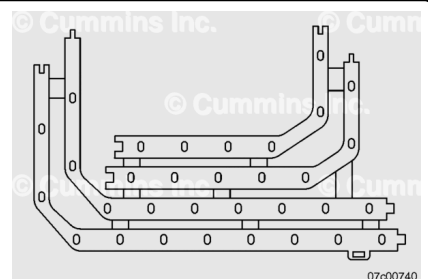
Assemble a new gasket onto the oil pan.

If using the silicone beaded gasket, place the oil pan gasket on the oil pan mounting flange with the words OIL PAN SIDE (1) facing the oil pan.



The replacement lubricating oil pan gasket may be in be a four piece segmented lubricating oil pan gasket rather than a single piece lubricating oil pan gasket.

The replacement lubricating oil pan gasket will need to be separated into four pieces and assembled on the lubricating oil pan flange to make a complete lubricating oil pan gasket.



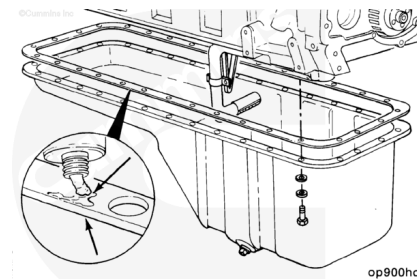
Apply a 2 mm [0.08 in] bead of sealant, Part Number 3164067, to the lubricating oil pan flange. Sealant must be applied around the circumference of the oil pan mounting holes.

Lay the lubricating oil pan gasket(s) onto the lubricating oil pan flange.

For segmented lubricating oil pan gaskets, make sure the lubricating oil pan gasket segment joints are properly aligned.

Apply a 2 mm [0.08 in] bead of sealant, Part Number 3164067, to the top of the new lubricating pan gasket.

For segmented lubricating oil pan gaskets, make sure the sealant fills the lubricating oil pan gasket segment joints.



⚠ CAUTION ⚠

Do not use the capscrew to pull the transfer tube into place during installation. Doing so will cause misalignment of the transfer tube, which can cause damage to the engine.

Be sure the transfer tube is aligned with the cylinder block suction tube hole when installing the oil pan.

Reach inside the engine. Push the transfer tube completely up into place in the cylinder block suction tube mounting hole by hand.

Install and tighten the capscrews in the sequence shown.

Torque Value: 54 n•m [40 ft-lb]

